

Rahul Rathnavel K

AI/ML Engineer

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SUMMARY

AI/ML Engineer with hands-on experience building AI-powered applications, backend systems, and IoT solutions through freelance and academic projects. Ranked in the Top 2% of the Amazon ML Challenge and as a Top 0.1% National Finalist in LTTS Techgium, with 850+ LeetCode problems solved. Experienced in machine learning, LLM/agentic workflows, REST APIs, and cloud/DevOps technologies.

TECHNICAL SKILLS

Programming Languages: Python, Java – **Databases:** PostgreSQL, MongoDB

Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD, Azure Pipelines, GitHub Actions

AI/ML & Tools: Machine Learning, LLMs, Agentic AI, Git, GitHub, Postman, Android Studio

Core CS: Object-Oriented Programming, Data Structures, Algorithms, DBMS, Operating Systems, Computer Networks, REST APIs, Version Control

EXPERIENCE

Freelance Software & AI Engineer — Independent Projects May 2024 – Present

- **Privacy-Preserving ML Pipeline:** Engineered custom CNN, LSTM, and RVFL models to analyze encrypted smart grid data, developing the core architecture for an award-winning academic project.
Tech Stack: Python, CNNs, LSTMs, RVFL, Machine Learning
- **Decentralized Blockchain Voting:** Built a secure electronic voting platform integrating smart contracts with Arduino-based biometric (fingerprint and facial) hardware authentication.
Tech Stack: React, Node.js, Python, C++, Solidity, Arduino IDE
- **Real-Time IoT Agriculture App:** Developed an Android application and low-latency API pipeline to process real-time Arduino telemetry, winning 1st Prize at a state-level hackathon.
Tech Stack: Java, Kotlin, REST APIs, Android Studio, IoT Hardware

KEY PROJECTS & ACHIEVEMENTS

- **Open Source Contributions (Keras, SciPy, Matplotlib, Statsmodels):** Contributed 5+ merged pull requests to major Python libraries, fixing bugs, improving functionality, and resolving merge conflicts through standard Git workflows.
- **Smartops (LTTS Techgium 9th Edition, Top 0.1%):** Architected an AI-powered autonomous incident engine for DevOps. Implemented agentic workflows and integrated LLM APIs (DeepSeek, Kimi) with Azure Pipelines to automate root cause analysis and log monitoring.
Tech Stack: Python, DeepSeek/Kimi APIs, Azure Pipelines, AWS DynamoDB, Agentic AI
- **Smart Glass Assistive System for Visually Impaired:** Built a real-time Flutter/Dart Android assistive application integrating on-device detection models. Designed and developed Python FastAPI REST APIs, custom YOLO, ViT, and CNN models.
Tech Stack: Flutter, Dart, C#, Python, SQLite, FastAPI, REST APIs, Git

CERTIFICATIONS

Oracle — [Cloud Generative AI Professional](#) [DeepLearning.AI](#) — [Deep Learning Specialization](#)
MongoDB — [Certified Associate Developer](#) Oracle — [Cloud Infrastructure DevOps Professional](#)

EDUCATION

Anna University Regional Campus, Coimbatore
B.Tech in Artificial Intelligence and Data Science

2022–2026
CGPA: 8.02/10